

**PMorph: A Task-Specialized Robot Manipulator
Morphology Optimizer**

by

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PMorph: A Task-Specialized Robot Manipulator Morphology Optimizer

Thesis directed by Prof. Alessandro Roncone

This paper seeks to address the energy-inefficiency, lack of task specialization, and overactuation seen in manufacturing from the current suite of generalized robotic manipulators. The approach involves specifying a task in a MuJoCo simulated environment and then using a PPO agent to optimize the morphology for that environment given a reward function. The morphological elements to be optimized include the number of links, the lengths of each link, and the type of joints involved. The reward function is tuned to weigh energy cost, end effector movement, end effector accuracy, manipulability measures, number of links, and task completion. PPO is ideal for this task because it can efficiently optimize over the continuous action space that is morphology, resulting in a robot that is more simple, efficient, and capable than those commonly used in industry today.

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Chapter 1

Introduction

Imagine a factory where every employee has to be trained on and work every job in the plant. Research ([46][30]) shows that approach to be inefficient because workers cannot build deep expertise and become very productive. The majority of factories today use specialized workers for this reason, dividing tasks boosts efficiency, reduces costs, and increases output [21][20]. And yet, this is exactly the opposite approach of robotics in manufacturing today where high costs and lack of tailorability are some of the most cited problems for the adoption of robotics [32][36]. To make matters worse [39] and [17] show that most robot manipulators are overactuated, inefficient, and ineffective for most tasks they are used in today.

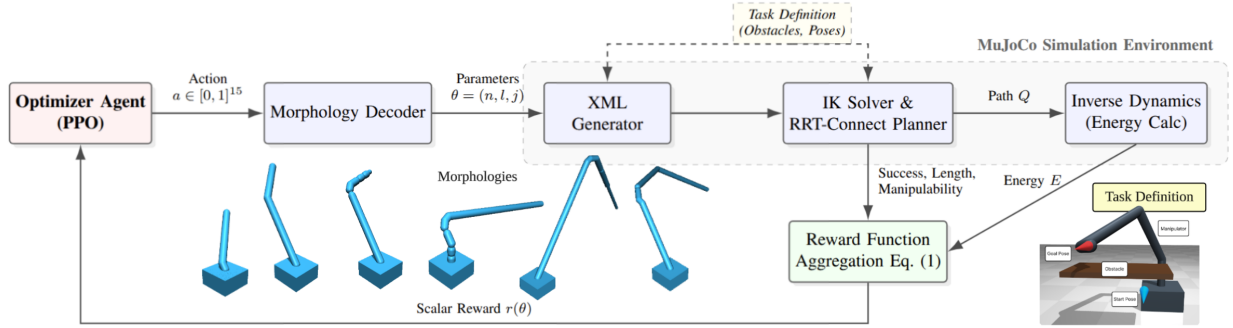


Figure 1.1: System overview of the proposed morphology optimization framework. The optimizer agent generates actions $a \in [0, 1]^{15}$, which are decoded into robot parameters $\theta = (n, l, j)$ (number of links, lengths, and joint types). These are used to create an XML model evaluated in a MuJoCo simulation environment, incorporating task definitions (obstacles and start/goal poses). IK solving and RRT-Connect path planning compute trajectories Q , followed by inverse dynamics for energy E . Performance metrics (success, path length, manipulability, and energy) are aggregated via the reward function in Eq. (1) to yield a scalar reward $r(\theta)$.

What if, in the same way that factories do with human workers, robots were specialized in the tasks they performed? Much like human workers physically and cognitively adapt to master repetitive tasks over time [3][31][54], robotic systems stand to gain significant efficiencies through morphological specialization. As advancements in additive manufacturing and Industry 4.0 drive the cost of mass customization closer to that of traditional mass production [53][4], it is becoming both economically feasible and operationally necessary to co-optimize a robot’s physical hardware alongside its control software for specific workspaces. This paper seeks to optimize the morphological design of robots to specific tasks by using a PPO agent to optimize the morphology based on various performance metrics and task-specific physical constraints. This ensures that the robot is as simple, efficient, and capable as possible for its particular job while still allowing the designer to specify the level of robustness.

PPO was chosen for its ability to explore the continuous action space that is morphology and it has been shown to do so efficiently and robustly [43]. The core method is to use Stable Baselines3 PPO implementation to iterate on the design parameters (number of links, joint types (X, Y, Z hinge, and Z actuation), and link lengths), use MuJoCo to simulate that iteration on a simple task using a simple controller (RRT-Connect and damped least squares (DLS) inverse kinematics (IK) for path planning), and feed the results (success, efficiency, and manipulability index) back into the PPO model. OMPL’s RRT-Connect was chosen for simplicity and it has been shown to be faster than other methods [26, 48] which enables faster and more robust training. DLS IK was chosen for a similar reason, it has been shown to be robust especially in situations where the target position is out of reach [6] which is inevitable when exploring morphologies, some of which will be too small or otherwise poorly suited to the task as the agent learns.

The evaluation signals will be successful path planning, path length, energy cost, and manipulability index. The manipulability index is a measure of a manipulators ability to arbitrarily change the position and orientation of the end effector at a particular point, borrowed from [56]. The tasks will be kept simple but varied in ways to test different types of movements like moving near the base, moving from near to far, different elevations, more linear movements, and different

obstacle constraints. One might imagine that a lot of pick and place tasks are mostly linear and so might largely be done by linear actuators more efficiently for example.

The research questions to be answered:

- (1) Can PPO be used to generate more efficient morphologies?
- (2) Can morphologies be generated for tasks that generalized manipulators cannot do or cannot perform well?
- (3) Can more simple morphologies (less DOF) be generated that can perform the same tasks with similar levels of accuracy?

Chapter 2

Related Work

While most of the work in this area has focused on the co-optimization of morphology and control for atomic, modular robots, AI has been increasingly shown to aid in the design of efficient robotics. This section dives into the background research specifically pertaining to the three research questions: energy efficiency, task specialization, and simplicity in manipulator design. Can PPO be used to do it?

2.1 Energy Efficiency

One of the primary motivations for morphology optimization is energy efficiency. Recent literature has shown that the traditional approach of optimizing controllers for energy efficiency can be significantly augmented by optimizing the morphology itself. [5] shows that PPO can be used to generate freeform morphologies from atomic modules for efficient gaits. [29] shows that Soft Actor Critic (SAC) can also be used to create more efficient gaits, this time by codesigning morphology and control on quadrupeds. [28] uses RL to do path planning for modular, reconfigurable, tiling robots that includes morphology configuration and is more energy efficient than conventional methods. This work provides the motivation to use PPO to optimize morphology for energy efficiency.

2.2 Task Specialization

Basic manipulators often struggle in highly constrained or atypical environment which prompts research into task-specific morphology design. [23] demonstrates the effectiveness of PPO and A3C

at generating configurations for reconfigurable tiling robots to maximize coverage in specialized environments. [8] proves that DDQN can be used to generate morphologies and controllers for reconfigurable robots in order to do tasks that fixed morphologies cannot or cannot perform well. [45] did the same thing with the addition of a foundation model to speed training and increase effectiveness. PMorph builds on this by using obstacle and task-aware morphology optimization to design around morphology weaknesses that basic arms would fall into.

2.3 Simplicity

FANUC 6-DOF series of manipulators are the most commonly used in industry today [37][47] and the 7-DOF Franka Emika Panda is considered a standard in research [12][35]. However, [39] and [17] indicate that these common, general-purpose manipulators are overactuated for most of the tasks they are used in leading to parasitic mass and computational waste. [9] and [34] show that reduced DOF designs can be efficient and effective. PMorph addresses simplicity directly by having a reward term that can be used to punish additional DOF, allowing optimization for accuracy and simplicity at the same time.

Chapter 3

Problem Formulation

The task is formulated as a single-step reinforcement learning problem in a MuJoCo simulated environment. A morphology θ is chosen, run through the simulation, and then evaluated based on weighted reward function terms.

3.1 Morphology Parameterization

A candidate morphology is defined by the tuple

$$\theta = (n, l, j),$$

where

- $n \in \{2, \dots, 7\}$ is the number of links,
- $l = [l_1, \dots, l_n] \in [0.05, 1.2]^n$ are the link lengths (m),
- $j = [j_1, \dots, j_n] \in \{0, 1, 2, 3\}^n$ are the joint types (0: revolute about x, 1: about y, 2: about z, 3: slide in z).

The action space of the RL agent is the continuous box

$$\mathcal{A} = [0, 1]^{1+7+7},$$

from which n , l , and j are decoded.

3.2 Task Specification

The robot has to move from start poses to goal poses (position + unit quaternion) in a cluttered workspace with fixed obstacles (walls, shelves, ceilings). An example of such an environment may be seen in Figure 3.1, which shows the container task where two sets of start and goal poses are set at either ends of a shipping-container-like workspace with the arm positioned in the middle. The poses are modified with Gaussian noise ($\sigma = 0.05$) at evaluation in order to encourage robust designs.

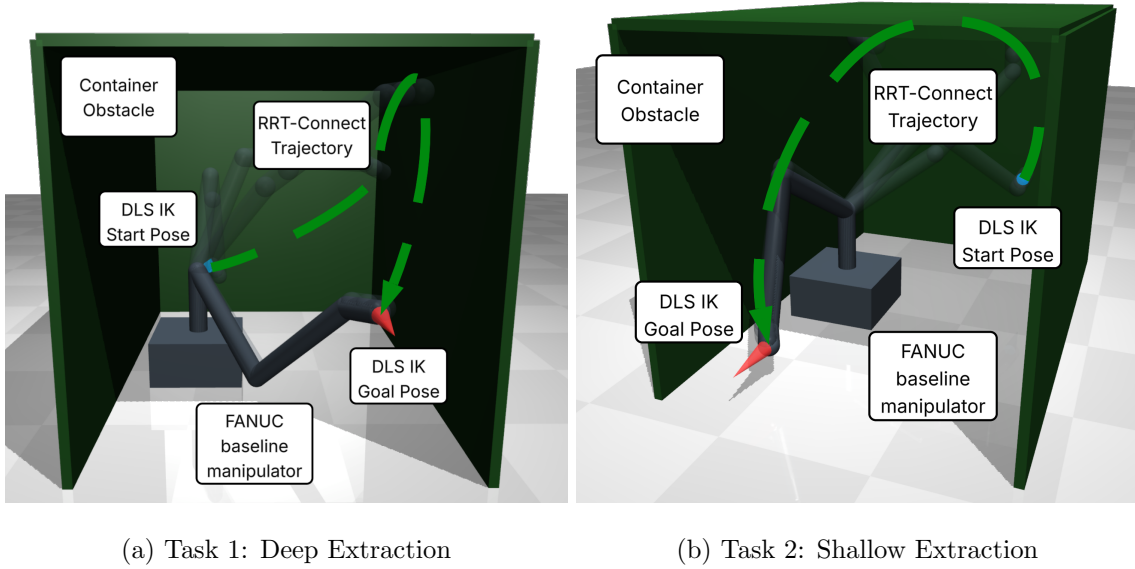


Figure 3.1: **Container task evaluation.** The manipulator is tasked with reaching deep into a constrained, container-like environment (start poses in blue) and extracting to an open area (goal poses in red). This highly constrained task tests the morphology’s compactness and ability to perform high-reach tasks without colliding with ceiling or wall constraints.

3.3 Evaluation Pipeline

Morphology θ is evaluated as follows:

- (1) A MuJoCo XML model is generated with θ and the task poses/obstacles.

- (2) Inverse Kinematics is solved for the start and goal poses using Damped Least Squares.
- (3) A path between the start and goal configurations is computed using RRT-Connect.
- (4) If a path connect be found, return reward $30 - 100 \times \text{IK error}$.
- (5) Otherwise, calculate the other reward terms and return the weighted sum:

- End-effector path length $p = \sum \|\mathbf{x}_{ee}^{t+1} - \mathbf{x}_{ee}^t\|$ (Euclidean distance between each point along path),
- Manipulability $\mu = \min(\sigma(\mathbf{J}))$ (minimum singular value of manipulability jacobian at start and goal [24]),
- Energy $E = \sum \|\boldsymbol{\tau} \odot \mathbf{v}\| \cdot \Delta t$ (MuJoCo inverse dynamics using constant velocity segments along path at torque limits).

3.4 Reward Function

The scalar reward is

$$\begin{aligned}
 r(\theta) = \frac{1}{N} \sum_{i=1}^N & \left[w_0 \cdot \text{success}_i - w_1 \cdot p_i \right. \\
 & - w_2 \cdot (e_{\text{start},i} + e_{\text{goal},i}) + w_3 \cdot (\mu_{\text{start},i} + \mu_{\text{goal},i}) \\
 & \left. - w_4 \cdot (n - 2) - w_5 \cdot E_i \right]
 \end{aligned} \tag{3.1}$$

where N is the number of start-goal pairs, e is IK error, success is valid path, and the weights (w_k) are tuned to balance the competing objectives (accuracy vs. efficiency vs. manipulability). The final optimization problem is therefore

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{\pi(\cdot|\phi)} [r(\theta)],$$

where π is the PPO policy given the parameters ϕ .

This formulation encodes the challenges outlined in Section 1:

- (1) Energy efficiency via the energy term.

- (2) Task specialization via the success, accuracy, and manipulability terms.
- (3) And simplicity via the link number and path length terms.

Chapter 4

Method

4.1 PPO Action Decoding

The PPO agent selects an action $a \in [0, 1]^{15}$ which is then decoded into morphology parameters θ as shown in Algorithm 1.

Algorithm 1 Morphology Decoding from PPO Action

- 1: **Input:** action $a \in [0, 1]^{15}$
 - 2: $n \leftarrow \text{round}(a_0 \cdot (n_{max} - n_{min}) + n_{min})$ ▷ links: 2–7
 - 3: $l \leftarrow a_{1:8} \cdot (l_{max} - l_{min}) + l_{min}$ ▷ lengths: 0.05–1.2m
 - 4: $j \leftarrow \text{round}(a_{8:15} \cdot 3)$ ▷ types: 0–3 (hinge X/Y/Z, slide Z)
 - 5: **Return:** $\theta = (n, l_{:n}, j_{:n})$
-

4.2 Simulation

θ is then passed into the evaluation environment, which is implemented as a Gymnasium class ‘robotArmEnv’.

- (1) **XML Generation:** A dynamic MuJoCo model is built via ‘generateXML(n, l, j)’, which assembles the links sequentially according to the parameter specifications using revolute/slide joints and capsule geoms (mass = l_i). Fixed obstacles (walls, ceiling, shelves, floor) are included as per the task set up.
- (2) **Path Planning:** Damped least-squares IK (robust variant with collision-aware line search)

solves for start/goal configurations. RRT-Connect then finds a collision-free joint-space path [26], using constant-velocity segments for energy estimation via MuJoCo inverse dynamics.

- (3) **Reward Computation:** $r(\theta)$ from Eq. (3.1) is returned with N being the number of sets of start and goal poses in the task and noise ($\sigma = 0.05$) added to poses for robustness.

4.3 Training

Training follows the standard PPO loop over 1M timesteps (single-step episodes):

Algorithm 2 PMorph Training

- 1: Initialize PPO policy π_ϕ
- 2: **for** each episode $t = 1$ to 1M **do**
- 3: Sample $\theta \sim \pi_\phi(\cdot|\phi)$
- 4: $r \leftarrow \text{eval}(\theta)$ ▷ Section 3 pipeline
- 5: Store (θ, r) in buffer
- 6: Update ϕ via the clipped surrogate loss [43]:

$$\begin{aligned} \mathcal{L}(\phi) = & \hat{\mathbb{E}}_t \left[\min(r_t(\phi)\hat{A}_t, \right. \\ & \left. \text{clip}(r_t(\phi), 1 - \epsilon, 1 + \epsilon)\hat{A}_t) \right] \\ & + S[\pi_\phi](s_t) - \beta \cdot \mathbb{E}_t[(\hat{V}_\phi(s_t) - V_t^{\text{targ}})^2] \end{aligned}$$

▷ $\epsilon = 0.2$, GAE advantages [42]

- 7: **end for**
-

Hyperparameters: $\epsilon = 0.2$ (clip ratio) and $\beta = 0.5$ (value loss weight)

4.4 Evaluation

While Proximal Policy Optimization (PPO) was the primary algorithm driving the morphological optimization in this study, the optimized morphology θ^* was also evaluated against alterna-

tive optimization strategies to validate the efficacy of the PPO approach. Specifically, PPO’s performance was compared against Soft Actor-Critic (SAC), also implemented via the Stable-Baselines3 library [38], and the Covariance Matrix Adaptation Evolution Strategy (CMA-ES), implemented using the standard `cma` Python library [15, 14]. After training, the optimized morphology θ^* is evaluated on a test set of start/goal pairs with noise. The reward terms are averaged over that test set to assess performance. A comparison is made using the baseline Panda and FANUC-like arm morphology. Based on the most commonly used model (FANUC LR Mate 200iD [2]), the baseline has 6 links, with lengths of 0.165, 0.330, 0.08, 0.285, 0.05, 0.05 (meters), and joint types of 2, 0, 0, 2, 0, 2 (revolute about z, x, x, z, x, z). The Panda is also approximated from its product manual [10] with 7 links, lengths of 0.333, 0.316, 0.0825, 0.0825, 0.384, 0.107, 0.05 (meters), and joint types of 2, 0, 2, 0, 2, 0, 2 (revolute about z, x, z, x, z, x, z).

Chapter 5

Experimental Setup

To evaluate PMorph, the MuJoCo (version 3.3.7) physics engine was used across a set of tasks specified below using the reward function defined in Equation (3.1) and the evaluation pipeline described in Section 3.

5.1 Tasks

5 tasks were chosen to test different types of motions (different axes, rotations, and positions w.r.t. the base) and obstacle constraints to create a thorough and practical evaluation. All tasks use a workspace centered at the robot base (0, 0, 0) except for the “Wall Mount” task where the base is moved to the middle of the near wall (0, -0.2, 0.6) and rotated $-\pi/2$ about the x-axis to face outward from the wall. The tasks were sized such that the start and goal poses fell within the dextrous workspace of the baseline models (approximately 0.35m to 0.5m from the base) in order to ensure that the comparison is as ideal as possible for baselines, except for the “Outreach” and “Wall Mount” tasks where the goal poses are near the reach limits in order to test research question 2 more effectively. [56] shows that a manipulator’s manipulability index approaches zero near the boundaries of the reachable workspace and operating in those low manipulability regions will artificially inflate energy costs and path lengths [52]. The tasks are as follows:

- (1) **Container Task:** The robot is positioned on the floor of a shipping-container-like environment in the middle and slightly to one side with walls and a ceiling (0.6m x 0.4m x

0.4m) all around and one open end. The start poses are near and facing the back of the container and the goal poses toward and near the open end (see Figure 3.1).

- (2) **Wall Mount Task:** The robot is mounted on the middle of a wall facing an opposing wall with a shelf mounted in the middle. One start pose is to the side of the mount wall facing down and the corresponding goal pose is on top of the shelf facing toward that wall. The other set is the opposite where the start pose is on the shelf facing the wall and the goal is to the side facing down (Figure 5.1).
- (3) **Shelf Task:** A single shelf obstacle is placed off to the side of the robot at 0.3m height. There are two sets of start and goal poses alternately above and below the shelf all facing in the positive x direction (Figure 5.2).
- (4) **Outreach Task:** There are no obstacles in this task, the robot is on an open floor, the start pose is near the base facing out and the goal pose is out near the limits of reach for the baseline models (Figure 5.3).
- (5) **Side-to-Side Task:** Similar to Outreach Task in that there are no obstacles and robot is on open floor, there are two sets of start and goal poses that are alternately on either side of and facing away from the robot (Figure 5.4).

The appendix contains the specifications for all of the task obstacles and poses.

5.2 Metrics

Performance is evaluated on each of the tasks averaged over 100 runs with fresh noise each time. Each of the reward terms and the total reward is reported for each task and morphology:

- Success rate: fraction of valid IK + RRT paths.
- Accuracy e (m and rads).
- Path length p (m).

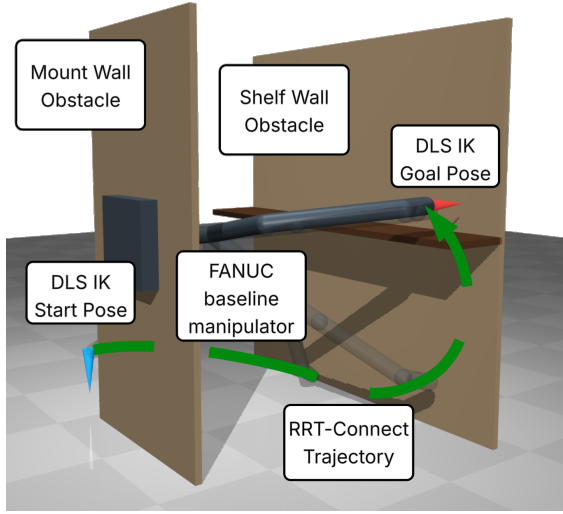
- Energy E (J).
- Manipulability μ (min singular value).
- Link count n .

5.3 Implementation and Compute

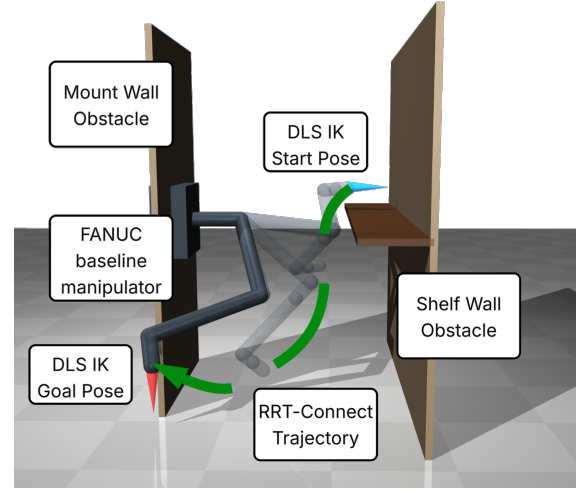
Training uses Stable Baselines3 PPO (1M timesteps, single-step episodes) on a 11th Gen Intel i9-11900KF CPU. The environment was vectorized with 24 parallel instances ('SubprocVecEnv'). Full hyperparameters are in Table 5.1.

Table 5.1: PPO Hyperparameters

Parameter	Value
Learning rate	0.001
Horizon (n_{steps})	256
Batch size	512
Epochs per update	4
Discount factor (γ)	0.98
Clip ratio (ϵ)	0.2
Value loss coefficient (β)	0.5
Entropy coefficient	0.0

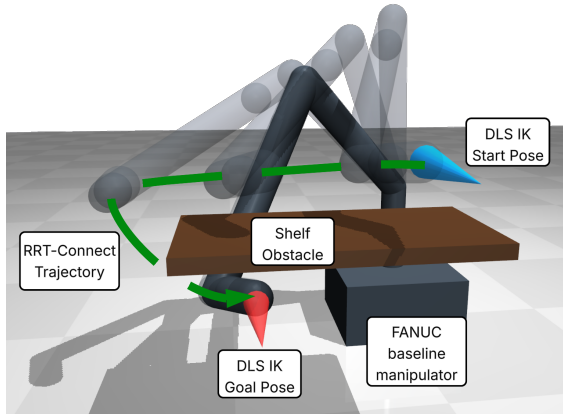


(a) Task 1: Mount Wall to Shelf

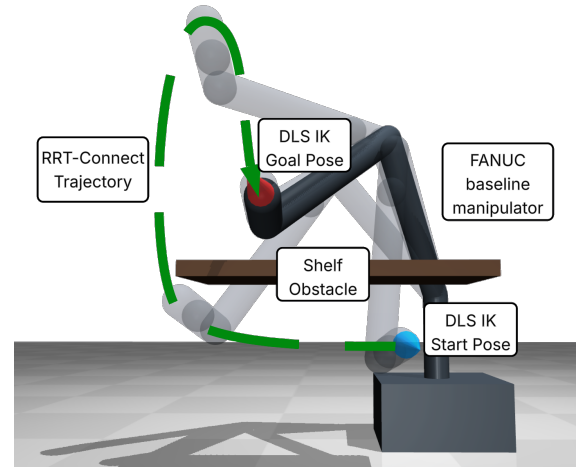


(b) Task 2: Shelf to Mount Wall

Figure 5.1: **Wall Mount task evaluation.** The manipulator, mounted horizontally on the near wall, must navigate to and from the opposing shelf. Faded overlays represent the RRT-generated trajectory. This configuration evaluates a morphology's ability to resist gravity in atypical base orientations while navigating constrained pick-and-place trajectories.



(a) Task 1: Top to Bottom



(b) Task 2: Bottom to Top

Figure 5.2: **Shelf task evaluation.** The manipulator navigates around a floating shelf obstacle, moving from a start pose (blue marker) to a goal pose (red marker). Transparent models illustrate the intermediate joint-space trajectory, concluding at the opaque target configuration. This task specifically tests the morphology's capability to maneuver through vertically constrained spaces and perform complex over-and-under reaching motions typical in industrial pick-and-place operations.

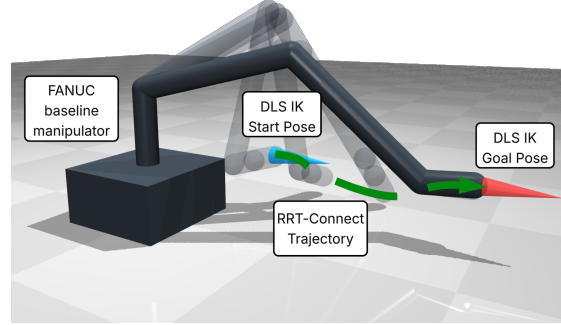
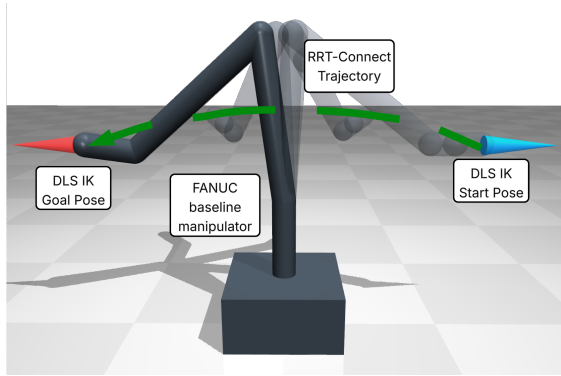
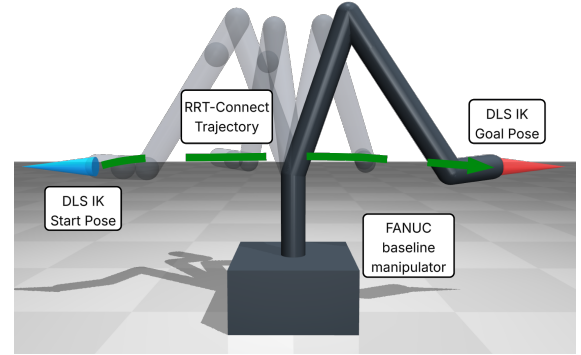


Figure 5.3: **Near-Base Outreach task evaluation.** The manipulator extends from a configuration near its base (blue marker) to a distant target coordinate (red marker). Transparent overlays depict the collision-free joint-space trajectory generated by RRT-Connect, terminating at the opaque goal configuration. This obstacle-free environment specifically isolates and evaluates a morphology’s maximum functional reach and extension efficiency.



(a) Task 1: Left to Right



(b) Task 2: Right to Left

Figure 5.4: **Side-to-Side outreach evaluation.** The manipulator navigates from the start pose (blue marker) to the goal pose (red marker) across the workspace. Faded models illustrate the intermediate trajectory states discovered by RRT-Connect. This task evaluates the morphology’s ability to maximize lateral reach and avoid collision with its own base.

Chapter 6

Results

Each optimized morphology θ^* found by PMorph was compared against the baseline FANUC and Panda across 100 runs in each task. The average total reward and individual reward terms are reported.

Main Table?

6.1 Energy Efficiency

6.2 Task Specialization

6.3 Simplicity

Figure 6.1 shows the SAC generated morphology for the container task and it is significantly more simple than the generic arms but also significantly less accurate.

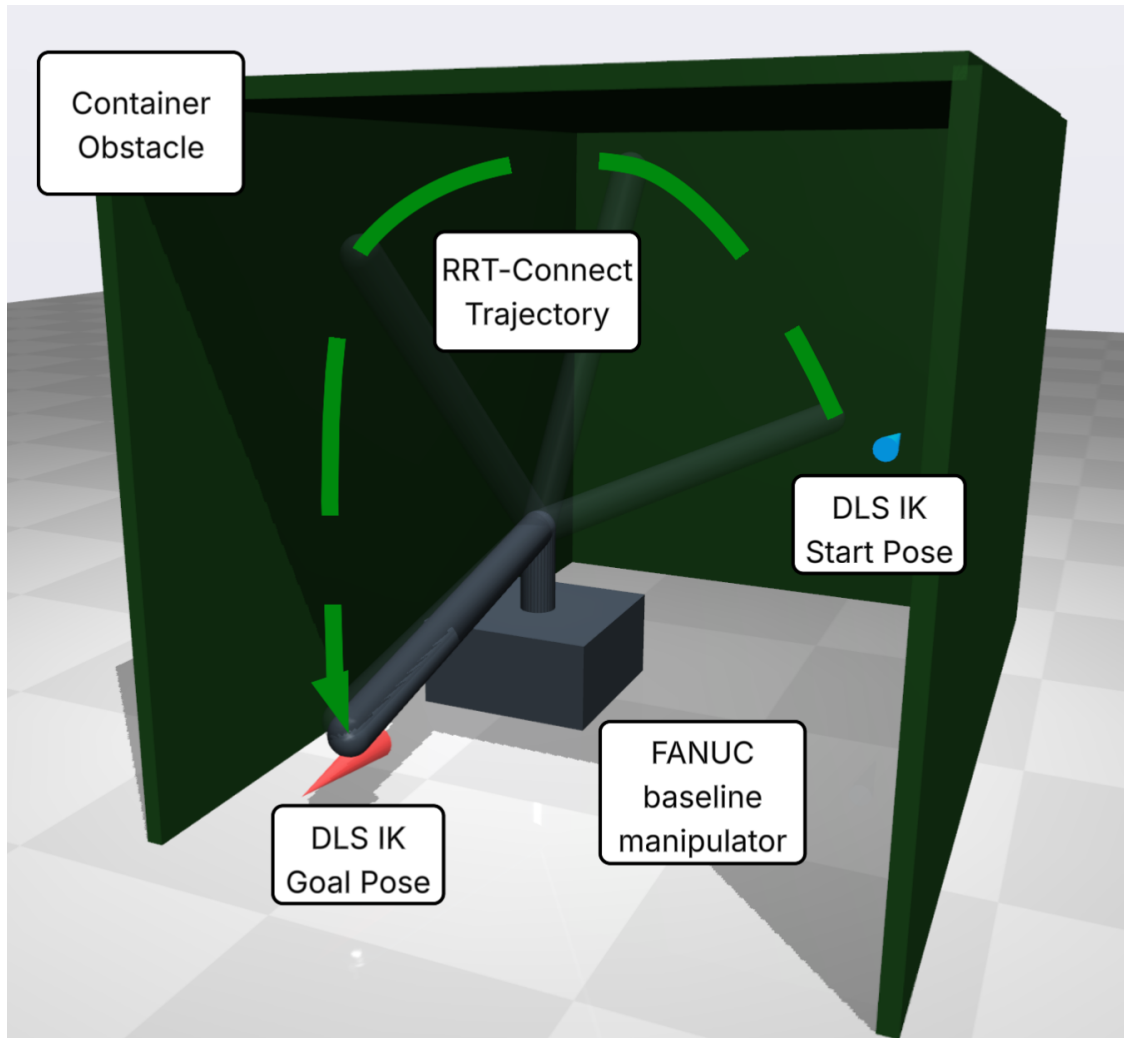


Figure 6.1: **SAC-generated Container Task Morphology.** The SAC-generated morphology is shown with a trajectory in the container task. The morphology is simple with 2 links, one z-hinge that's 0.1179m and one y-hinge that's 0.5151m. The manipulator is not quite long enough to reach the start position and significantly off the orientation, but appears right on the goal pose.

Chapter 7

Conclusion

The results were promising at first and remain so in simple tasks, but as the tasks become more constrained the simple controller starts failing to find paths, especially for higher link number arms which biases the agent towards selecting simpler arms which can't score highly but can consistently get paths.

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Appendix A

Task Specifications

Table A.1: Container Task Poses

Name	Position	Quaternion
Start 1	$[-0.45, 0.0, 0.2]$	$[0.7071, 0, -0.7071, 0]$
Goal 1	$[0.45, 0.4, 0.3]$	$[0.7071, 0, 0.7071, 0]$
Start 2	$[-0.45, 0.4, 0.3]$	$[0.7071, 0, -0.7071, 0]$
Goal 2	$[0.45, 0.0, 0.1]$	$[0.7071, 0, 0.7071, 0]$

Table A.2: Container Task Obstacles

Name	Position	Size
Back Wall	$[-0.6, 0.2, 0.4]$	$[0.01, 0.4, 0.4]$
Left Wall	$[0, -0.2, 0.4]$	$[0.6, 0.01, 0.4]$
Right Wall	$[0, 0.6, 0.4]$	$[0.6, 0.01, 0.4]$
Ceiling	$[0, 0.2, 0.8]$	$[0.6, 0.4, 0.01]$

Table A.3: Wall Mount Task Poses

Name	Position	Quaternion
Start 1	[-0.4, -0.2, 0.2]	[0, 1, 0, 0]
Goal 1	[0.5, 0.4, 0.7]	[0.7071, -0.7071, 0, 0]
Start 2	[-0.2, 0.4, 0.7]	[0.7071, -0.7071, 0, 0]
Goal 2	[0.4, -0.2, 0.2]	[0, 1, 0, 0]

Table A.4: Wall Mount Task Obstacles

Name	Position	Size
Mount Wall	[0, -0.2, 0.6]	[0.3, 0.01, 0.6]
Shelf Wall	[0, 0.6, 0.6]	[0.6, 0.01, 0.6]
Shelf	[0.0, 0.5, 0.6]	[0.6, 0.1, 0.01]

Table A.5: Shelf Task Poses

Name	Position	Quaternion
Start 1	[0.45, -0.1, 0.4]	[0.7071, 0, 0.7071, 0]
Goal 1	[0.35, -0.3, 0.2]	[0.7071, 0, 0.7071, 0]
Start 2	[0.35, -0.1, 0.2]	[0.7071, 0, 0.7071, 0]
Goal 2	[0.45, -0.3, 0.4]	[0.7071, 0, 0.7071, 0]

Table A.6: Shelf Task Obstacles

Name	Position	Size
Shelf	[0.4, -0.2, 0.3]	[0.1, 0.2, 0.01]

Table A.7: Near-Base Outreach Task Poses

Name	Position	Quaternion
Start	[0.2, 0.2, 0.12]	[0.7071, 0, 0.7071, 0]
Goal	[0.65, 0.2, 0.12]	[0.7071, 0, 0.7071, 0]

Table A.8: Side-to-Side Task Poses

Name	Position	Quaternion
Start 1	[0, -0.4, 0.3]	[0.7071, 0.7071, 0, 0]
Goal 1	[0, 0.4, 0.3]	[0.7071, -0.7071, 0, 0]
Start 2	[0, 0.4, 0.4]	[0.7071, -0.7071, 0, 0]
Goal 2	[0, -0.4, 0.4]	[0.7071, 0.7071, 0, 0]